MatryoshkaLoRA: Learning Accurate Hierarchical Low-Rank Representations for LLM Fine-Tuning

Vienna Deep Learning Meetup #76

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Vienna Deep Learning Meetup

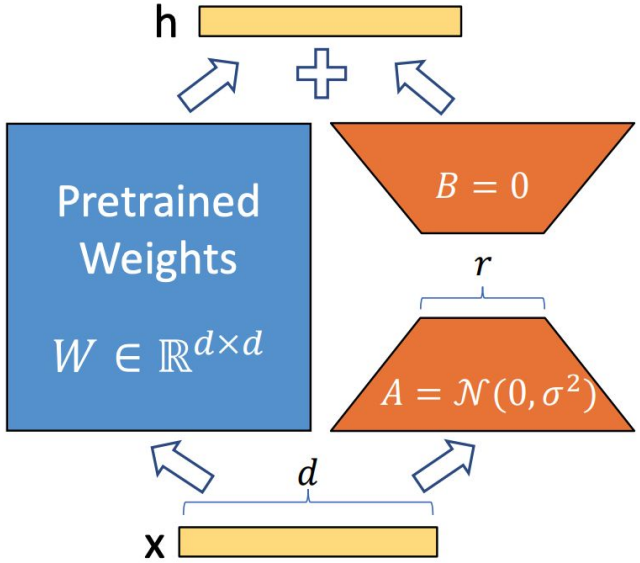
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Agenda

- LoRA fine-tuning
  - Intro
  - Advantages
  - Challenges
- Matryoshka Representation Learning
  - brief overview
- MatryoshkaLoRA
  - Formulation
  - Evaluation Metric
  - Results

LoRA fine-tuning - intro

- Pretrained model: $\mathbf{W} \in \mathbf{R}^{m \times n}$
- Define a rank **r**, with $\mathbf{r} \ll \min(m, n)$
- Learn adapters $\mathbf{A} \in \mathbf{R}^{m \times r}$ and $\mathbf{B} \in \mathbf{R}^{r \times n}$
- Keep **W** frozen and train only **A** and **B**
- Model at inference:
  - $\mathbf{W}_{\mathbf{LoRA}} = \mathbf{W} + \mathbf{AB}$



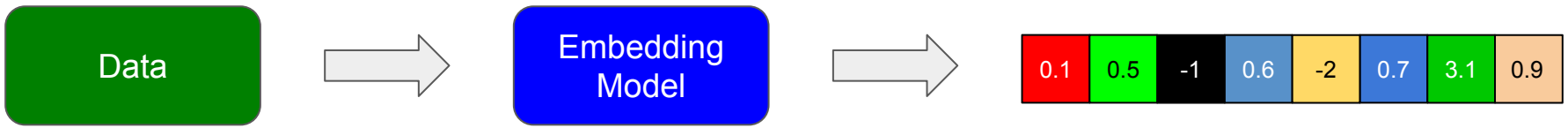
LoRA fine-tuning

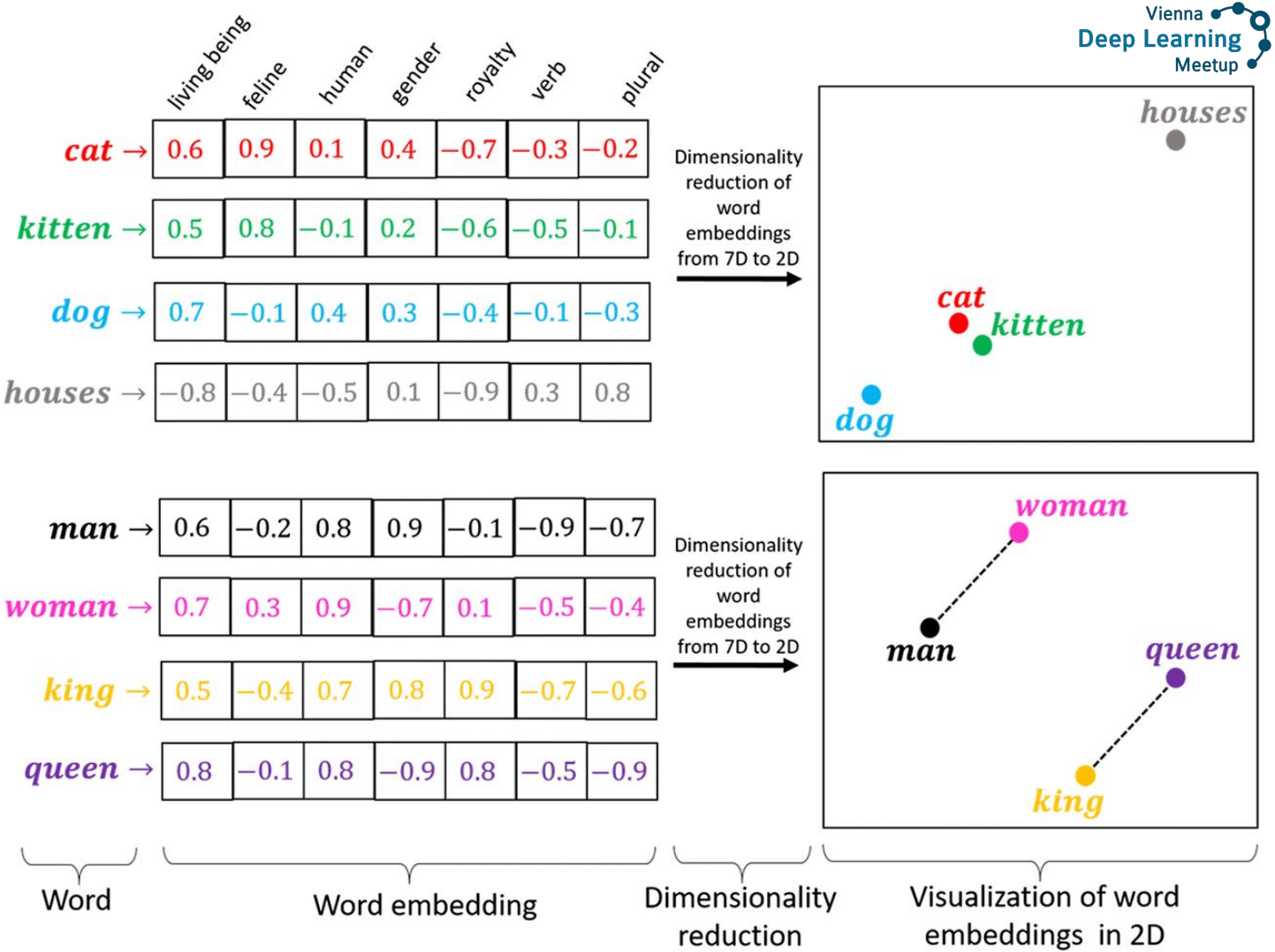
- Advantages
  - Few total trainable parameters: a fraction of **mn** when **r** is small (usually 1-2%)
  - Still pay the memory cost of large model for inference (**AB** can be added to **W**)
  - Can serve different specialized adapters at the same time

- Challenges
  - tuning the rank **r**: trade-off between performance and number of parameters (training time)
  - fine-tuning is still expensive for super large models
  - for a specific **r**: changing the rank requires retraining

Matryoshka Representation Learning

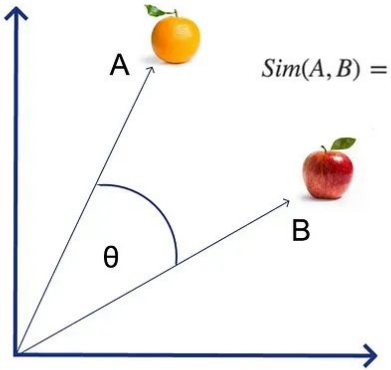
- Embeddings (general properties)
  - Numerical representations (vectors) of images, video, text, audio, etc.
  - Fixed size (usually a power of 2)
  - Useful for in LLMs to encode words (tokens), vector databases (RAG)
  - Automatically learned in LLMs, auto-encoders




living being
feline
human
gender
royalty
verb
plural
cat → 0.6 0.9 0.1 0.4 −0.7 −0.3 −0.2
kitten → 0.5 0.8 −0.1 0.2 −0.6 −0.5 −0.1
dog → 0.7 −0.1 0.4 0.3 −0.4 −0.1 −0.3
houses → −0.8 −0.4 −0.5 0.1 −0.9 0.3 0.8
Dimensionality reduction of word embeddings from 7D to 2D
houses
cat
kitten
dog
man → 0.6 −0.2 0.8 0.9 −0.1 −0.9 −0.7
woman → 0.7 0.3 0.9 −0.7 0.1 −0.5 −0.4
king → 0.5 −0.4 0.7 0.8 0.9 −0.7 −0.6
queen → 0.8 −0.1 0.8 −0.9 0.8 −0.5 −0.9
Dimensionality reduction of word embeddings from 7D to 2D
woman
man
queen
king
Word
Word embedding
Dimensionality reduction
Visualization of word embeddings in 2D

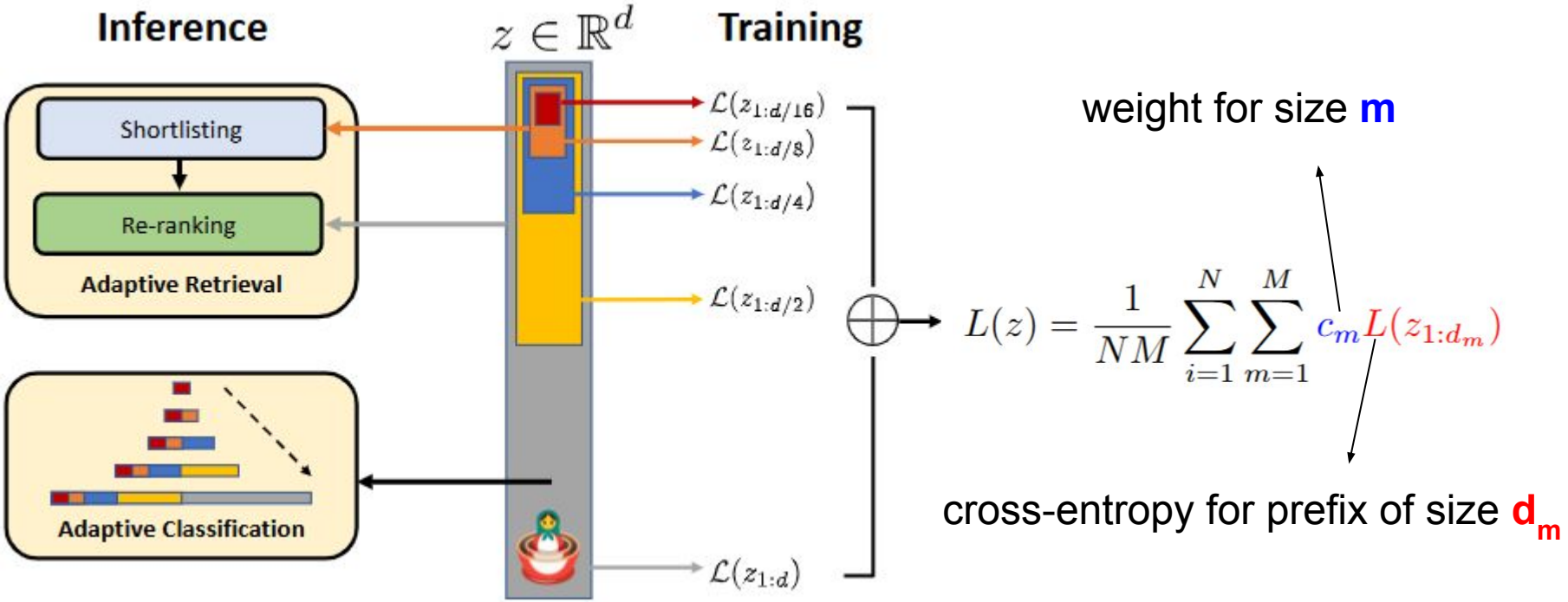
Matryoshka Representation Learning

- Measuring similarity in the embedding space
  - Cosine similarity
  - Expensive at large scale
- How to choose the embedding size?
  - Trade-off between expressivity and latency
  - Short embeddings: fast, less informative
  - Long embeddings: slower, more informative
- Best of both worlds:
  - Matryoshka Representation Learning
  - Learn prefixes that live in the same vector!



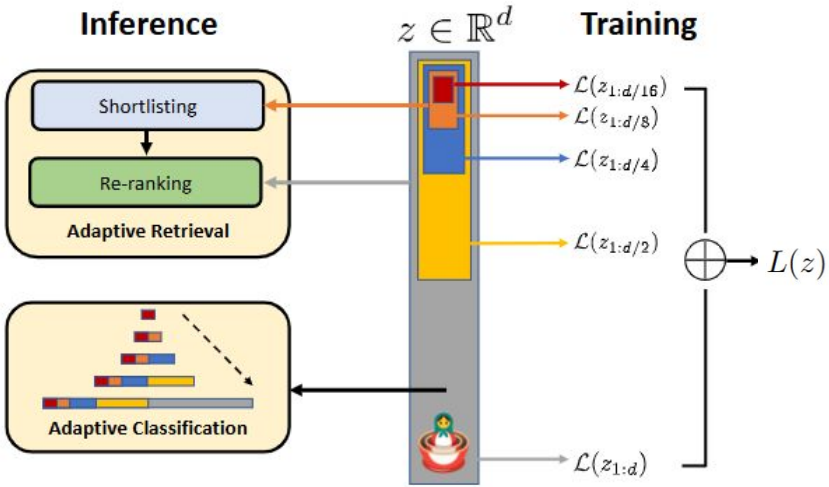


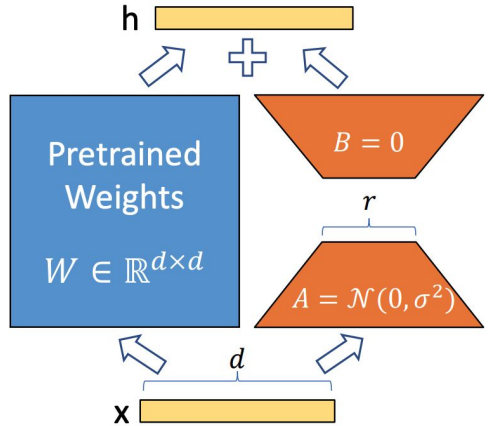
Matryoshka Representation Learning



Matryoshka LoRA

- How to combine Matryoshka and LoRA?
  - Learn prefixes of different ranks inside the same adapters **A** and **B**
- Purpose:
  - Use any rank after training once





Matryoshka LoRA - Formulation

- Matryoshka Representation Learning loss:

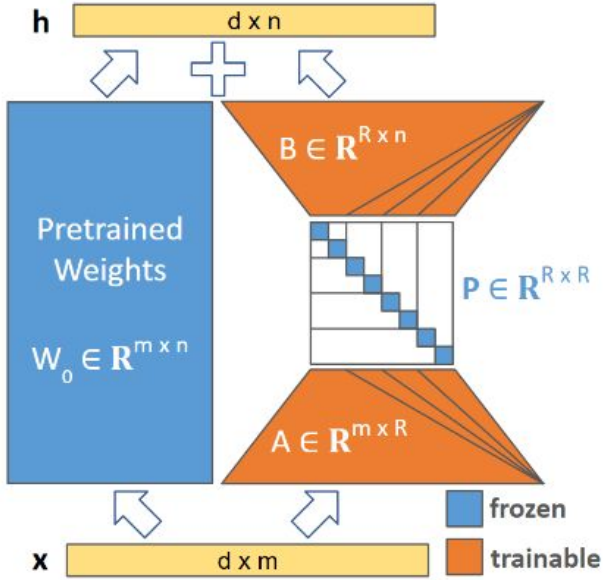
$$L(z) = \frac{1}{NM} \sum_{i=1}^{N} \sum_{m=1}^{M} c_m L(z_{1:d_m})$$

- LoRA inference:

$$W_{\text{LoRA}} = W + AB$$

- Matryoshka LoRA inference:

$$W_{\text{M-LoRA}} = W + \sum_{m=1}^{M} A_{1:d_m} B_{1:d_m}$$
$$= W + APB$$



Matryoshka LoRA - Formulation

- How does matrix **P** look like?
  - Suppose we have rank 8 and choose powers of 2: $\mathbf{d_m \in \{1, 2, 4, 8\}}$

$$W_{\text{M-LoRA}} = W + \sum_{m=1}^{M} A_{1:d_m} B_{1:d_m}$$
$$= W + APB$$

$$A_{1:1}B_{1:1} \rightarrow P_1 = [1,0,0,0,0,0,0,0]$$
$$A_{1:2}B_{1:2} \rightarrow P_2 = [1,1,0,0,0,0,0,0]$$
$$A_{1:4}B_{1:4} \rightarrow P_4 = [1,1,1,1,0,0,0,0]$$
$$A_{1:8}B_{1:8} \rightarrow P_8 = [1,1,1,1,1,1,1,1]$$

- Efficient implementation:
  - scale rows of **A** with elements of **P**
  - cheap operation

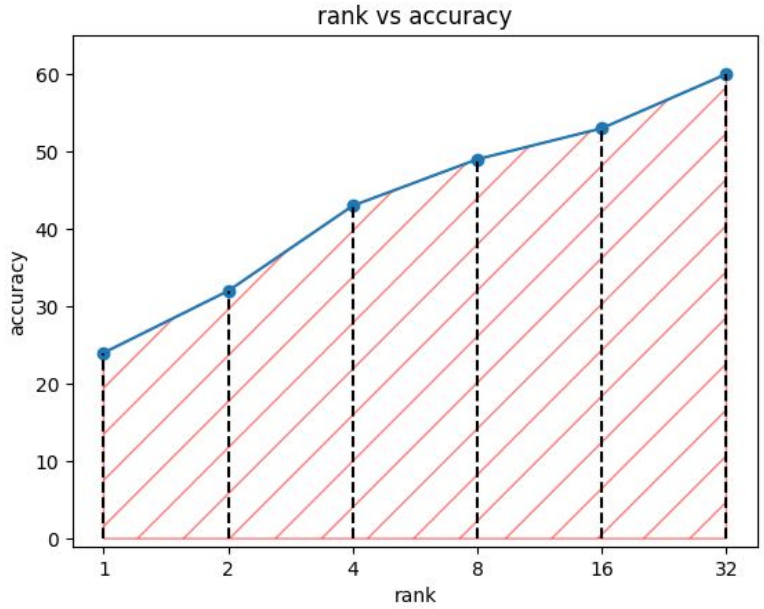
$$W_{\text{M-LoRA}} = W + (A * P)B$$

$$P = P_1 + P_2 + P_4 + P_8$$
$$= [4,3,2,2,1,1,1,1]$$

Matryoshka LoRA - Evaluation Metric

- How to choose the best model when we evaluate it on multiple ranks?
  - we created **AURAC**: **A**rea **U**nder the **R**ank **A**ccuracy **C**urve
  - unified metric to directly compare models when evaluated on multiple ranks
  - trapezoidal rule to compute hatched area under the rank-accuracy curve

$$\text{AURAC} = \frac{1}{r_M - r_1} \sum_{i=1}^{M-1} \frac{a(r_i) + a(r_{i+1})}{2} \cdot (r_{i+1} - r_i)$$



Matryoshka LoRA - Results

- Train Llama-3 model with 1B and 8B on GSM-8k
  - evaluation heavy: tested on all ranks that are power of 2 → **$\log_2(r)+1$** evaluations in total

- Comparison with LoRA:
  - standard slicing of adapters

- Emphasize that our scaling is effective

Matryoshka LoRA - Results

- GSM-8k dataset

  Question:
  Weng earns $12 an hour for babysitting. Yesterday, she just did 50 minutes of babysitting. How much did she earn?

  Answer:
  Weng earns 12/60 = $<<12/60=0.2>>0.2 per minute. Working 50 minutes, she earned 0.2 x 50 = $<<0.2*50=10>>10. #### 10

- N-shot prompting:
  - add N examples (Question, Answer) in the prompt
  - add Question
  - expect answer

Matryoshka LoRA - Results

- Llama-3-1B on GSM-8k (8-shot evaluation)
  - pre-trained model: 34.7%

Adapter Type	Eval Accuracy per Rank						AURAC
	1	2	4	8	16	32	
LoRA	33.3 ±1.1	34.3 ±.5	34.3 ±.2	34.3 ±.5	34.5 ±.2	34.6 ±.6	34.3
MatryoshkaLoRA	**35.5** ±.7	**36.1** ±.4	**37.4** ±1.1	**38.5** ±.6	**39.4** ±.2	**38.4** ±.9	**37.7**

Matryoshka LoRA - Results

- Llama-3-8B on GSM-8k (3/8-shot evaluation)

N-Shots PT	Adapter Type	Eval Accuracy per Rank						AURAC
		1	2	4	8	16	32	
3-shot	LoRA	74.1	74.5	74.2	74.3	74.1	73.8	74.2
PT: 74.0	Matryoshka	**74.8**	**76.9**	**77.3**	**77.6**	**77.5**	**77.6**	**77.1**
8-shot	LoRA	**77.8**	77.9	77.5	77.6	77.0	76.9	77.5
PT: 77.8	Matryoshka	77.6	**78.6**	**79.6**	**80.1**	**79.9**	**79.8**	**79.4**

Shoutout to my collaborators



Mher Safaryan



Dan Alistarh

Thank you!

Questions?

Resources

- Papers:
  - LoRA: https://arxiv.org/pdf/2106.09685
  - Matryoshka Representation Learning: https://arxiv.org/pdf/2205.13147
  - MatryoshkaLoRA: https://arxiv.org/pdf/2605.07850

- Images from:
  - https://medium.com/@hari4om/word-embedding-d816f643140
  - https://python.plainenglish.io/mastering-cosine-metrics-a-data-scientists-essential-toolkit-3ce95eda91f9?gi=3b78853e1e88
  - https://m.media-amazon.com/images/I/719GMeE5t-L._AC_SL1000_.jpg